DETAILED SOLUTION APPROACH DESIGN

Comprehensive overview addressing Initial concerns:

Key Sections:

1. **Current State Assessment**
   - Active sources (UKG, Salesforce, Ensora)
   - Paused sources (NetSuite, Coupa, Workable)
   - POC systems (CollaborativeMD)
2. **The Problem Statement**→Why we need governance & semantic layer
   - 5 critical problems: Inconsistent metrics, duplicate effort, governance gaps, technical debt, compliance risk
   - Business impact: Each leadership sees different revenue numbers, Board deck conflicts, analyst productivity loss
3. **The Solution: 5-Layer Architecture**

   Layer 5: AI/BI Semantic Layer (Innovation)
   ↑
   Layer 4: Operational Marts (Domain-Specific)
   ↑
   Layer 3: Core Data Warehouse (Star Schema)
   ↑
   Layer 2: Integration Layer (Reconciliation)
   ↑
   Layer 1: RAW Staging (Source Mirrors)

Deep dive into the AI/BI Mart layer (Semantic Layer):

Key Sections:

1. **What Is a Semantic Layer?**
   - Business-friendly abstraction layer
   - Single source of truth for metrics
2. **Why Traditional BI Fails**
   - **Before→** Every analyst writes custom SQL, Result: Two different "Revenue" numbers!
   - **After (Semantic Layer)→**Revenue defined ONCE
3. **Technical Architecture**

   Consumption Layer (BI tools, NL interface, APIs)
   ↓
   Semantic Layer Engine (dbt, MetricFlow, Git)
   ↓
   Operational Marts (Finance, HR, Sales, Ops)
4. **Metrics Catalog Structure**
   - Business definition, calculation logic, data sources
   - Dimensions, filters, metadata
   - Access control, quality checks, usage stats
5. **Natural Language Query Interface (AI-Powered)**
   - User: "Show me top 10 customers by revenue last quarter"
   - AI parses intent → queries semantic layer → generates viz

6. **Access Control & Data Governance**
    - **Role-Based Access Control (RBAC)**
        - Executive: Full access to all metrics
        - Finance: Full financial, read-only HR/Sales
        - Sales reps: Only THEIR deals (row-level security)
    - **Audit Logging:** Every query logged (who, what, when)

Comprehensive governance ensuring KPI consistency:

Key Sections:

1. **What Is Data Governance?**
    - Framework ensuring data is accurate, consistent, secure, compliant, trusted
    - **Without:** Different revenue numbers, no ownership, compliance risk
    - **With:** Single source of truth, accountability, audit-ready
2. **Governance Operating Model (4 Tiers)**

    Tier 1: Governance Council (Strategic)
    Tier 2: Data Domain Owners (Tactical)
    Tier 3: Data Stewards (Day-to-Day)
    Tier 4: Data Engineering Team (Technical)

3. **6 Data Governance Policies**
    Policy 1: Metric Definition Standard →Every metric must have: definition, calculation, source, owner, quality rules
    Policy 2: Data Quality Standards→Automated daily checks, alerts if threshold violated
    Policy 3: Access Control Policy→Sensitive data tiers (Public → Confidential → Restricted → Highly Restricted)
    Policy 4: Change Management Policy→All changes tracked in Git with approval workflow
    Policy 5: Data Retention Policy
    Policy 6: Compliance Policy
4. **Data Quality Framework (6 Dimensions)**
    - Completeness, Accuracy, Timeliness, Consistency, Uniqueness, Validity
    - Quality scorecard by domain
5. **Metrics Catalog Governance**
    - Lifecycle: Proposed → In Development → In Testing → Certified → Deprecated → Retired

AI/BI Mart Layer & Semantic Layer: Detailed Architecture & Implementation Plan

Current Reality: Raw data in Snowflake from 7 sources (UKG, Salesforce, Ensora active; NetSuite/Coupa/Workable paused; CollaborativeMD POC) .

Target: Build **AI/BI Mart Layer** (Operational Marts) + **Semantic Layer** enabling:

- **BI Use Cases:** Self-service dashboards, consistent KPIs for customers/endusers
- **AI Use Cases:** ML feature store, predictive models, natural language analytics

Architecture Layers (Snowflake-Native):

Semantic Layer	← Natural language, APIs, consistent KPIs
AI/BI Mart Layer	← Domain marts: Finance, HR, Sales, Ops
Integration Layer	← System reconciliation
Raw Layer	← Source mirrors (UKG, SALESFORCE etc.)

Real Limitations & Challenges

1. Data Availability Gaps (Critical Blockers)

2. Technical Debt & Data Quality Issues

RECONCILIATION CHALLENGES

COMPLEX MAPPINGS

3. Governance & Adoption Risks

ORGANIZATIONAL CHALLENGES:

• No Data Governance Council
• Analysts resist semantic layer over SQL
• Department silos

TECHNICAL RISKS:

• Snowflake costs → Semantic layer caching must be perfect
• ETL sync failures → Single point of failure

<u>**Detailed Technical Architecture**</u>

Layer 1: Raw Layer (ETL Syncs)

Purpose: Immutable source history, reprocessing capability

Layer 2: Integration Layer (System Reconciliation)

Purpose: Single master records across systems

Layer 3: AI/BI Mart Layer (Domain-Specific)

Purpose: Optimized for BI queries + AI feature engineering

Layer 4: Semantic Layer (Metrics + AI Query Engine)

Purpose: Business-friendly metric catalog + natural language interface

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Use case Building on AI/BI Mart + Semantic Layer Architecture

AI ready use cases as an example with current information

Use Case 1: Finance AR Intelligence

Business Requirements

Revenue Dashboard:

* AR aging buckets
* Top 10 risky customers (AI scored)
* Payment probability by customer

Self-Service:

* "Collections pipeline by collector"
* "DSO by customer segment"
* "Predict which invoices pay next week"

Data Flow:

```
NetSuite → INTEGRATION.dim_customer → MART_FINANCE.fact_ar_aging
                   ↓
           Semantic: dso(), ar_aging_bucket()
                   ↓
          Cortex ML: payment_probability()
```

Use Case 2: Procurement Intelligence

Business Requirements

Procurement Dashboard:

* Spend by category/supplier
* Invoice exception rate (56% → target 20%)
* Savings opportunities (AI identified)

Self-Service:

* "Spend trend by department"
* "Top 10 risky suppliers"
* "Invoice exceptions by reason"

Data Flow:

```
Coupa.Invoice → INTEGRATION.dim_vendor → MART_OPS.fact_procurement
                   ↓
           Semantic: spend_by_category()
                   ↓
          ML: supplier_risk_score()
```

NOTE: The above mentioned design approach and use cases are based on current market use cases and the data sources documentation shared along with our analysis, there may be deviations going further based on the nuances of the environment and data.